Integration — Lesson 22 Test: Sigma Notation & Riemann Sums

10 questions · show your work in the box · label left/right/midpoint clearly · justifications need full sentences

Name: ______________________________ Date: ______________ Score: ______ / 10

1. Evaluate: $\Sigma_{k=1}^{6}$ k. 1 pt

2. Evaluate: $\Sigma_{k=1}^{3}$ (k^2 + 1). 1 pt

3. Write in sigma notation (do not evaluate): 3 + 6 + 9 + 12. 1 pt

4. For $f(x) = x^2$ on [0, 3] with n = 3 equal slices, compute the LEFT Riemann sum and the RIGHT Riemann sum. 1 pt

5. For $f(x) = x^2$ on [0, 2] with $n = 2$ equal slices, compute the MIDPOINT Riemann sum. (First find Δx and the midpoint of each slice.) 1 pt

6. Water flows into a tub at rates measured every 4 minutes: t (min): 0, 4, 8, 12 · R(t) (L/min): 6, 9, 7, 5. Estimate the total liters over the 12 minutes with a LEFT Riemann sum. 1 pt

7. A cyclist's velocity is measured at uneven times: t (s): 0, 1, 4, 6, 10 · v(t) (m/s): 8, 6, 5, 7, 4. Estimate the distance traveled on [0, 10] with (a) a left sum and (b) a right sum. (Compute each subinterval's width first!) 1 pt

8. $f(x) = \sqrt{x}$ is increasing on [1, 4]. Does a RIGHT Riemann sum overestimate or underestimate $\int_1^4 \sqrt{x}\,dx$? Justify your answer in a full sentence (name the behavior of f and connect it to the rectangle heights). 1 pt

9. R(t) is the rate, in liters per minute, at which juice is poured into a vat, where t is measured in minutes. Using correct units, explain what $\int_0^{20} R(t)\,dt$ represents. 1 pt

10. Water leaks from a tank at the measured rates: t (hr): 0, 3, 5, 9 · r(t) (gal/hr): 12, 10, 8, 6. (a) Estimate the total gallons leaked over [0, 9] with a left Riemann sum. (b) Estimate it with a right Riemann sum. (c) r(t) is decreasing; which estimate is the overestimate? Justify. 1 pt